

Kazi Md. Tawsif Rahman

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Research Summary

Computer Science graduate and backend systems engineer studying the privacy and detection trade-offs of DP-SGD for network intrusion detection. My independent study uses Opacus privacy accounting, a five-seed NSL-KDD comparison, a supplementary single-seed UNSW-NB15 check, and shadow-calibrated membership-inference attacks. The analysis is complete; a venue-neutral manuscript draft is under author review and has not been submitted.

Research Interests

AI security; differential privacy; privacy-preserving machine learning; membership-inference auditing; trustworthy evaluation of security-critical ML; federated learning for security applications; reproducible experimental systems.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh July 2024
B.Sc. in Computer Science and Engineering CGPA: 3.59 / 4.00

Research Experience

Privacy–Utility Auditing of DP-SGD for ML-Based Network Intrusion Detection

Independent Research Project 2026 – Present

- Designed a locked NSL-KDD binary-classification protocol with a 70/10/20 target-train/validation/shadow split, validation-only F2 threshold selection, and untouched KDDTest+ utility evaluation.
- Completed a condition-matched five-seed comparison of a non-private PyTorch MLP and Opacus DP-SGD at actual $\epsilon \approx 4$ and 2. Recorded δ , clipping, sampling, noise, epochs, and accountant settings. The earlier $\epsilon \approx 8$ sweep remains single-seed context and is not part of the five-seed comparison.
- On KDDTest+, five-seed mean Recall was 70.93% non-private, 71.34% at $\epsilon \approx 4$, and 70.11% at $\epsilon \approx 2$. The $\epsilon \approx 4$ Recall difference was uncertain; its false-positive rate rose from 5.61% to 8.76%, while average precision fell from 93.75% to 89.07%.
- Evaluated score-only and label-aware membership-inference attacks trained and calibrated on condition-matched shadow models. Overall attack ROC-AUC was near 0.50 even without DP, and paired intervals did not support a measured overall leakage reduction. The DP guarantee is conditional on fixed, non-private preprocessing.
- Completed a supplementary single-seed UNSW-NB15 validation under the same three model conditions. Selected-threshold F1 stayed near 0.85, while private models had slightly higher false-positive rates and lower average precision. This check does not establish universal cross-dataset generalization.

Code, results, and reproducibility artifacts: github.com/tawsif113/privacy-utility-dp-ids

Publications

- Kazi Md. Tawsif Rahman** and Chowdhury Mahfuzul Hoq. “An Automated System for Detecting Property Insurance Fraud Using Machine Learning.” *International Journal of Mathematical Sciences and Computing*, 2024. doi:10.5815/ijmsc.2024.03.02.
- Mahfuzulhoq Chowdhury, **Kazi Md. Tawsif Rahman**, and Hossain Ahmad Maruf. “Whistle Blower: An Insurance Awareness Mobile Application with Insurance Policy Selection, Fraud Detection, Critical Help, Complaint Features.” *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, pp. 1–6, 2024. doi:10.1109/CSITSS64042.2024.10817002.

Selected Professional Experience

Software Engineer, BRAC IT Services Ltd., Dhaka, Bangladesh

Dec 2024 – Present

CRM Platform

- Built configurable rule-based routing with priorities, schedules, conditional matching, rotation rules, duplicate checks, and validated field/operator/value conditions.
- Replaced full-column Kanban reordering with fixed-length order keys and reduced each movement to a single-rank assignment.
- Added access controls, idempotent moves, transition validation, audit logs, and timeline entries to make state changes traceable.

Ticketing System

- Built services for ticket creation, search, comments, forwarding, assignment rules, teams, SLA/KPI tracking, feedback, notifications, and metadata.
- Implemented least-loaded team routing with Redis-backed workload counters and maintained PostgreSQL–Redis workload consistency after ticket creation, reassignment, and resolution.

Technical and Research Skills

Research methods	Locked-split experimental design, validation-only threshold selection, shadow-model membership-inference auditing, privacy accounting, paired confidence intervals, reproducibility manifests, threat-model specification
ML / Privacy	Python, scikit-learn, XGBoost, PyTorch, Opacus, DP-SGD, membership-inference attacks, model evaluation
Backend / Data	Java, Spring Boot, Spring Security, JPA/Hibernate, REST APIs, PostgreSQL, MongoDB, Redis, RabbitMQ, Flyway
Engineering	SQL, C, C++, Docker, Linux, Git, JUnit 5, Mockito, API testing

Awards and Achievements

- Codeforces Specialist; rating above 1500.
- Finalist, CUET IUPC 2024 and CUET IUPC 2023.
- Finalist, CUET Tech Carnival 2020.

Selected Coursework

Algorithms and Data Structures; Database Systems; Computer Networks; Operating Systems; Software Engineering; Artificial Intelligence; Machine Learning; Information Security.